Exam topics #11-20

11. The principle of the lock in amplifier. Phase locked loop, PLL synchronization to external reference signal. Measuring a small signal in a noisy environment using a modulation technique. Noise filtering considerations – why should we use a lock-in? Higher harmonic measurements with a lock-in.
12. Experimental definition of noise. The spectral density of noise, the Fourier transform of the current-current correlation function and the Fourier transform of the current fluctuations, and the relationship between these quantities. Spectral density calculation based on DFT.
13. Thermal noise calculation, relation between conductance and noise through the correlation time. Minimum input noise of a current amplifier. Cross correlation measurement technique.
14. Shot noise of independently emitted electrons, electron charge measurement. Equivalent noise bandwidth. Resistance fluctuations, 1/f type noise. Aliasing in noise measurement, antialiasing filter.
15. International System of Units (SI). Old and new definitions of basic units, reasons for the introduction of the new SI system. Historical evolution of time and distance measurement.
16. Atomic clock as a time standard. The quantum metrology triangle: converting voltage to time through the Josephson effect, current to voltage through the quantized Hall effect and current to time by electron pumps. Mass measurement: Watt balance and Avogadro project.
17. Magnetic field sensors: inductive sensors; magnetoresistive sensors (anisotropic magnetoresistance, giant magnetoresistance and spin valve sensors, tunneling magnetoresistance), Hall probes, SQUIDs.
18. Distance and position sensors: inductive sensors, capacitive sensors, touch screen with pressure sensitive pen, laser rangefinders and ultrasonic distance sensors, LIDAR system.
19. Temperature sensors, primary and secondary thermometers. Thermocouples, resistance thermometers, thermistors. Light and electromagnetic radiation sensors: photodiodes, CCD sensors, CMOS active pixel sensors, bolometers. Accelerometers: MEMS accelerometers and gyroscopes, piezoelectric accelerometers.
20. Measurement of radiation and particles. Interactions of ionizing radiation and matter, neutral particle detection. Gas-filled ionization detectors: ion chambers, proportional counters and Geiger-Müller tubes. Scintillation detectors with photoelectron multipliers. Applications. Fundamentals of nuclear electronics, integral and pulse mode detection, energy resolved spectra.

3

Introduction – Measurement Techniques

What are measurement techniques? Well, it means different things in every profession.

From a physicist's point of view: we usually want to measure a **physical quantity**. Often, a **sensor** with a special physical principle converts that physical quantity into an electronic signal. The accuracy and success of the measurement, however, depends largely on how the sensor signal is processed, what **measurement circuit**, what **measurement setup**, what measurement trick is used. Finally, data are acquired with a precision **measuring instrument**, and various **data evaluation** tricks and data science methods can also be important for processing these data. A measurement technique course may cover almost any of these aspects.

Physical quantity	Sensor	Measurement setup, measuring circuit	Measuring instrument	Data processing

Topics covered:
-Measuring superconductivity: an experiment illustrating measurement difficulties and tricks. How could we refine the measurement? The concept of feedback-controlled temperature-regulation: the PID control.
-Modern multimeters, functions generators, DAQ cards, oscilloscopes: the concept of A/D and D/A converters, typical device specifications and tricky measurement modes. Related basics of amplifier circuits.
-Suppression of external disturbance signals: elimination of capacitive and inductive pick-ups by shielding and twisted cables, offset-compensation of thermoelectric voltages, guarding.
-High-frequency measurements: RC-time-constant issues, wave propagation in transmission lines.
-Spectrum analysis, the effect of finite measurement time and sampling rate, window functions, Fast Fourier Transformation (FFT), various types of spectrumanalyzers.
-Phase sensitive detection: lock-in amplifiers and phase-locked loops.
-Natural noise sources in high precision measurements, concept of noise analysis.
-Metrology, the SI system and measurement standards.
-Sensorics: measuring displacement, temperature, light, magnetic field, acceleration, etc.
-Introduction to nuclear measurement techniques

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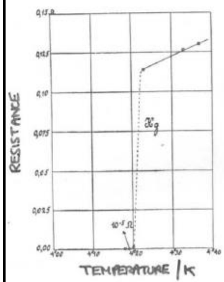
Measuring the zero resistance of a high-temperature superconductor sample – superconductivity in a nutshell

Discovery (1911):
The resistance of mercury drops to zero (unmeasurably small) below T=4.15 K



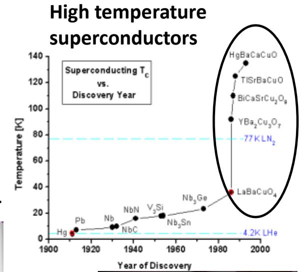
Heike Kamerlingh Onnes

Nobel prize: 1913

Microscopic theory (BCS theory, 1956)

Nobel prize: 1972
Schrieffer Bardeen Cooper

High temperature superconductors



K. Alex Müller and Georg Bednorz

Nobel prize: 1987

Further Nobel prizes:

Landau 1962

Abrikosov and Ginzburg 2003

Josephson and Giaever 1973

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Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

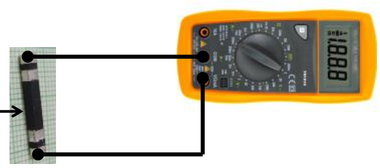
The **resistance** of a superconductor **should drop to zero below a** certain **critical temperature**.
First, we should look up the **sample specifications**: what is the material, what are the dimension, what is the critical temperature, and the maximum allowed current.

CSB-2/3/20 SUPERCONDUCTING BAR
Width: 3 mm
Thickness: 2 mm
Net weight: 0.001 kg
Material: Bi-2223
Length: 20 mm
Max. current: 60mA
Critical temperature: ~108K

How can we measure the resistance of the superconducting rod? Let's try with a digital multimeter!

Physical quantity to be measured: Resistance

superconducting sample



Do we trust the results? No, we are measuring the resistance of the cables and the contacts in addition to the resistance of the superconducting rod.

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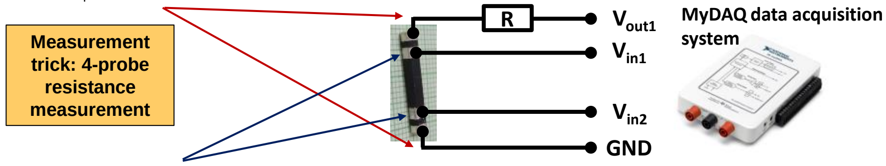
Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

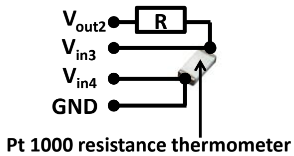
How can we refine the resistance measurement?

$I_{sample}=V_{out1}/R$ current is passed through the sample



Meanwhile the $V_{sample} = V_{in1}-V_{in2}$ voltage drop is measured on two different contacts from the current injection contacts. The voltage measurement is performed by a voltmeter of almost infinite internal resistance, i.e. practically no current flows in these voltage measurements lines, and no voltage drops on these cables and contacts. Therefore, V_{sample} only includes the true voltage drop on the superconducting rod, and $R_{sample}=V_{sample}/I_{sample}$ is the true sample resistance.

On the other side of the sample holder the temperature is measured by a Pt 1000 resistance thermometer in a “quasi four probe” fashion, meaning that there are only two contacts, but four wires, i.e. the resistances of the voltage measurement wires is cancelled.



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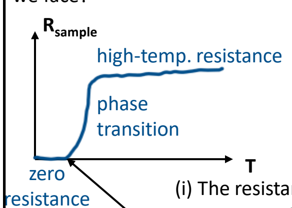
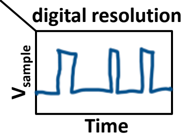
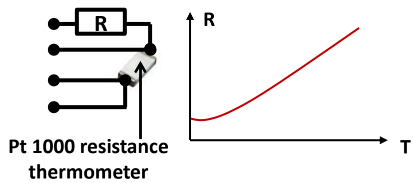
Measuring the zero resistance of a high-temperature superconductor sample – resistance measurement

Let’s perform the measurement. We do see the superconducting transition, but what kind of difficulties do we face?



(i) The resistance resolution of the measurement is restricted by the **digital resolution** of the card

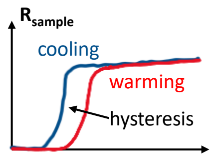




(ii) Due to the thermometer’s **nonlinear resistance vs. temperature characteristics** at low temperatures we may not trust the measured transition temperatures

(iii) The transition is different when the sample is cooled (blue) or warmed (red), i.e. a hysteresis is observed. Reason: the thermometer and the sample are not in thermal equilibrion, e.g. the themrometer cools and warms faster than the sample.

SOLUTION: well-controlled slow cooling and warming with a **PID temperature controller** (see later).



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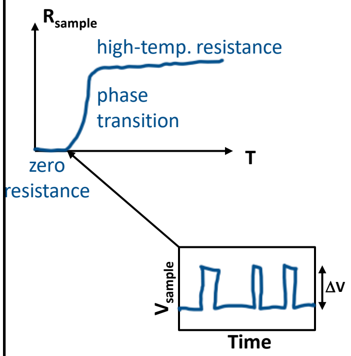
Measuring the zero resistance of a high-temperature superconductor sample

How accurate is the measurement?

20mm x 3mm x 2mm



Resolution of the four-probe resistance measurement according to the digital voltage measurement error:

Measuring current: I_{sample}=50mA

Digital resolution of the voltmeter: ΔV=0.01mV

Resolution of the resistance measurement: ΔR= 0.2mΩ

Dimensions: h=20mm, A=2mm x 3 mm

-> Resistivity resolution: $\Delta\rho$= ΔR A/h=6 x 10^{-8} Ωm

As a comparison, the resistivity of copper at room temperature: $\rho_{Cu,T=300K}$ =1.7 x 10^{-8} Ωm!!!

We do see a sudden transition, but the error of measuring zero resistance is larger than the resistance of a similarly sized copper rod would be.

Can we use a completely different measurement method to reduce the resistance measurement error by several orders of magnitude?

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Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement

Measurement trick: measuring circulating superconducting current by a magnetometer

If we could put **current** to a **superconducting ring**, it would **circulate forever** due to the zero resistance. This persistent current is easily **measured by a magnetic field sensor**.

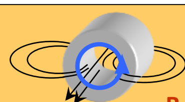
How can we put this current to the ring? Let us cool down the superconductor in external magnetic field. Inside the ring the Φ=BA flux changes in time as:

$$-\frac{\mathrm{d}\Phi}{dt} = V = RI = 0$$

Where R is the resistance of the ring, A is its cross section, and B the magnetic field inside. In case of zero resistance the flux cannot change inside, i.e. once the external field is switched off, a circulating current should arise in the ring, which maintains the original magnetic field inside the ring. This way it is easy to establish a circulating persistent current.

The superconducting ring is cooled in external magnetic field

Once the field is switched off a circulating current starts which maintains the original field



How fast would the current decay in an a ring with finite resistance?
(L is the ring's self-inductance and τ is the characteristic decay-time)

$$RI = -\frac{\mathrm{d}\Phi}{dt} = -L\frac{\mathrm{d}I}{\mathrm{d}t} \rightarrow B(t) \sim I(t) = I_0 e^{-\overbrace{(R/L)}^{1/\tau} t}$$

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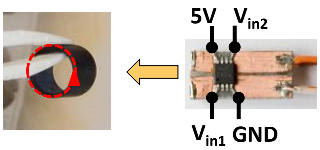
Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement

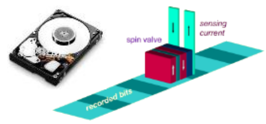


GMR magnetic field sensor

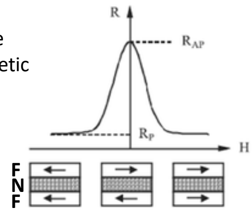
A. Fert és P. Grünberg

Nobel prize: 2007.

Applied sensor: The persistent current of the ring can be measured by a Giant Magneto Resistance (GMR) sensor.



GMR (Giant MagnetoResistance) phenomenon: electrons pass through such a nanostructure which consists of a nonmagnetic (N) layer between two ferromagnetic (F) layers. The resistance is sensitive for the relative orientation of magnetization in the two F layers.



This phenomenon is used in the reading head of spinning hard disk drives. The magnetization direction of one F layer is fixed, while the direction of the other is set by the magnetization of the disk (this is called spin valve).

Explanation: The magnetic layers polarize the spin of electrons passing through. The arrangement is similar to optical polarizers: light can only pass through two polarizers of the same orientation, while it cannot pass through them if the orientations are perpendicular. The GMR phenomenon occurs only in nanostructures if the two magnetic layers are very close to each other.

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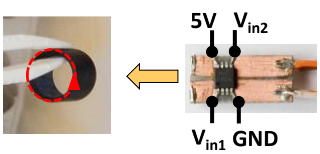
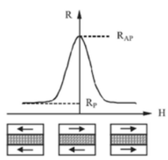
Measuring the zero resistance of a high-temperature superconductor sample – persistent current measurement



GMR magnetic field sensor



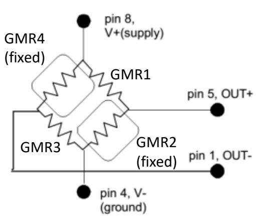
Instead of a single sensor, 4 similar GMR sensors are applied in a bridge arrangement. In two of these (GMR2 and GMR4) the antiparallel magnetic orientation of the layers is fixed, it cannot be changed by the external field.

Voltage is applied at two opposite corners of the bridge (pins 4 & 8) and voltage is measured at the other two corners (pins 5 & 1).

If the field is zero, but the temperature changes, the resistance of the 4 sensors changes in the same fashion, i.e. the bridge remains balanced, and the output voltage difference remains zero.

If a magnetic field is applied, the fixed GMR sensors (GMR2 and GMR4) keep their resistance, while the resistance drops for the working GMR sensors (GMR1 and GMR3), and thereby the unbalanced bridge outputs a finite voltage, which is proportional to the strength of the magnetic field.

Measurement trick: bridge arrangement



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Measuring the zero resistance of a high-temperature superconductor sample
resistance vs. persistent current measurement

Height: h=15mm
Diameter: d=15mm
Wall thickness: w=1.5mm

We have seen that our four probe method measures the zero resistivity with $\Delta\rho = 6 \times 10^{-8}\ \Omega$m error due to the finite digital resolution. What would be the τ current decay time constant in our superconducting ring, if its resistivity was not zero, rather the above $\Delta\rho = 6 \times 10^{-8}\ \Omega$m resistivity error value?

$\tau = L/R$, $R = d\pi\ \rho/(h\ w) = 0.125\mathrm{m}\Omega$, $L = \mu_0 A/h = 4\pi \times 10^{-7} \times d^2\pi/(4h) = 1.5 \cdot 10^{-8}$ H -> $\boldsymbol{\tau = 120\mu s}$ **than the four-probe resistance measurement.**

In case of persistent current measurement, even if the magnetic field measurement is somewhat unstable, we may say, that the current decays by less than 10% within a time of t=10minutes=600s.
-> $I(t) = I_0 \exp(-t/\tau) > 0.9 \cdot I_0$ -> $\boldsymbol{\tau > 6000s}$

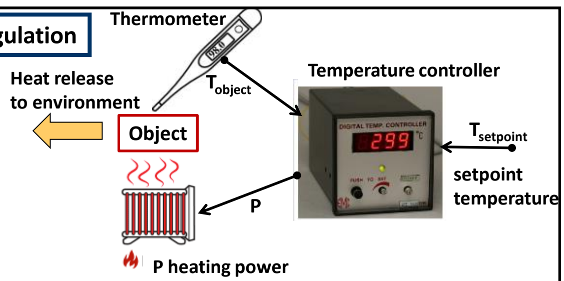
This calculation shows, that **the persistent current measurement determines the zero resistance with almost 8 orders of magnitude better resolution than the four-probe resistance measurement.**

The magnetic field of a persistent solenoid magnet in a Nuclear Magnetic Resonance (NMR) spectrometer was measured for several years. From this the current decay time-constant of the superconductor turned out to be $\tau > 10^5$ years, with the corresponding resistivity $\rho_{sc} < 10^{-32}\ \Omega$m!!
J. File & R. G. Mills, Phys. Rev. Lett 10, 93 (1963)

13

Precise temperature control: PID regulation

A simple control task: an object is to be heated to a given $T_{setpoint}$ target (setpoint) temperature by a heating element. The temperature (T) is measured and the heating power (P) is controlled based on the difference between the measured temperature and the setpoint temperature.



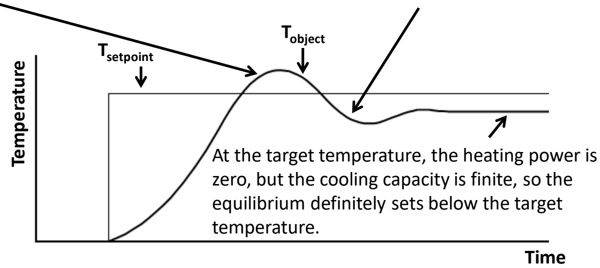
Simplest implementation: the operation of a room thermostat:
$$P = \begin{cases} P_0, \text{if } T_{object} < T_{setpoint} \\ 0, \text{if } T_{object} > T_{setpoint} \end{cases}$$

More elaborate, **proportional regulation:**
$$P = \begin{cases} C_p \cdot (T_{setpoint} - T_{object}), \text{if } T_{object} < T_{setpoint} \\ 0, \text{if } T_{object} > T_{setpoint} \end{cases}$$

If the object cools below the target temperature, the heating is switched on in proportion to the temperature difference, but the system does not react fast enough and cools below the target temperature.

Example: the target temperature is raised, how the control reacts:

As the object heats up above the target temperature, the heating is turned off, but the object continues to heat up due to the heat inertia.



If the C_p is too high, the overheating-overcooling oscillation becomes stationary, and no equilibrium is reached.

If the C_p value is too low, we cannot reach the target temperature at all.

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